

INDEPENDENT PEER-REVIEW PACKET

P02-LM-001

Conditional Degree-Four Deterministic Capture Majorant

Cold Review Packet 16 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	16 of 20
Object ID	P02-LM-001
Object class	Dynamical-systems lemma
Exact statement	For the exact pin-preserving degree-four vector field, REG_r and $s=q/r \leq -(2 K R^5 + 2R)$ imply the selected M-ward unstable branch from S enters the invariant strip, crosses centrally, enters an invariant contracting M-ball, and converges exponentially to M.
Internal terminal label	CLOSED - CONDITIONAL EXACT DEGREE-FOUR DETERMINISTIC CAPTURE MAJORANT
Terminal date	2026-07-24
Recommended expertise	Planar dynamical systems / invariant regions / hyperbolic equilibria

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Complete exact proof, derivation-blind organizationally distinct review, definition crosswalk, and 200,000-sample hostile falsifier.
Prior review record	Anthropic independently reconstructed the deterministic proof; shared-definition exposure was disclosed.
Binding exclusions	No exact-field transfer, Gaussian/Palm mass, adjacency rate, P0.1/P0.2, or Theorem B.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.
- Object-specific exclusions: No exact-field transfer, Gaussian/Palm mass, adjacency rate, P0.1/P0.2, or Theorem B.

4. Object-specific hostile checks

- Verify branch orientation at S and entry into the cone.
- Check inward flux on every strip and capture boundary.
- Verify central drift, finite-time transit, and exponential convergence without using probability.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

LCR-CAP-009-v1.0 — P02-LM-001 CLOSED CONDITIONAL DETERMINISTIC CAPTURE LEMMA PEER-REVIEW CAPSULE AND TERMINAL RECORD

Artifact ID: LCR-CAP-009-v1.0
Lemma ID: P02-LM-001
Prepared and adjudicated by: ChatGPT — OpenAI GPT-5.6 Thinking, Lemma Closure Director
Date: 2026-07-24
Class: CAP / CLS / EXACT DETERMINISTIC LEMMA / REVIEW CAPSULE
Authority: terminal lemma record within exact conditional deterministic scope under OP-PROT-011
Canonical theorem impact: NONE unless separately promoted by Dylan Roy
Terminal label: CLOSED — CONDITIONAL EXACT DEGREE-FOUR DETERMINISTIC CAPTURE MAJORANT

[[OBJ:P02-LM-001]] [[STATUS:CLOSED-CONDITIONAL]] [[SCOPE:DEGREE-FOUR-DETERMINISTIC-ONLY]] [[P0.2:OPEN]]
[[EXACT-FIELD:NOT-INCLUDED]] [[PROBABILITY:NOT-INCLUDED]]

1. READ FIRST

One-sentence result:

For the exact pin-preserving degree-four local vector field defined below, the explicit hypotheses $rR(U)^5 \leq 1/(1024K)$ and $q/r \leq -[2KR(U)^5 + 2R(U)]$ force the selected M-ward unstable branch of the saddle S to enter an invariant strip, traverse it, enter an invariant contracting neighborhood of M, and converge exponentially to M.

Why it matters:

This closes the deterministic finite-jet core of the P0.2 adjacency-to-one program. It replaces several formerly separate saddle-exit, strip, transit, and sink-capture obligations with one explicit polynomial-growth sufficient condition. It does not close P0.2 because exact-field transfer and probability integration are separate lemmas.

Current status:

CLOSED — CONDITIONAL EXACT DEGREE-FOUR DETERMINISTIC CAPTURE MAJORANT.

Authoritative proof:

GP-DER-047-v1.0, Sections 2–9.
Drive ID: 1jm1WxNmDd_rBLZA9FQsc2Nlzb9_PPCuEBV0fX3mCL1c.

Independent review:

CL-AUD-049-v1.0 — Anthropic/Claude derivation-blind analytic reconstruction.
Drive ID: 1fzL4HgJDsawb6ew66XpOCfJGnnxw6TFTom7BM3H8OnI.

Adversarial computation:

LCR-DATA-010-v1.0 — reconstructable 200,000-sample hostile stress capsule.
 Drive ID: 1It6VYebSawBCA3S7Zs_g7ndB26-yr_NH6K_yNQs373Q.

Recommended review order:

1. Exact statement and exclusions in Sections 2–4 of this capsule.
2. GP-DER-047 Sections 2–3 for the field and geometry.
3. GP-DER-047 Sections 4–9 for the analytic proof.
4. CL-AUD-049 Sections 1–5 for the derivation-blind reconstruction and lineage disclosure.
5. LCR-DATA-010 for hostile edge testing and source identity.
6. Closure and reopening rules in Sections 12–14 below.

2. FROZEN EXACT STATEMENT

Let

$U=(a,w,z,c_{40},c_{31},c_{22},c_{13},c_{04})$ in $\mathbb{R}^8$,

$R=R(U)=1+|a|+|w|+|z|+|c_{40}|+|c_{31}|+|c_{22}|+|c_{13}|+|c_{04}|$.

Fix

$K=4096$,

$\epsilon_0=1/(1024K)$.

Let $0 < r \leq 1$ and let $s=q/r$ be a real parameter. Define the pin-preserving degree-four vector field $F_r^{(4)}=(F_X,F_Y)$ on $\mathbb{R}^2$ by

$F_X(X,Y)$

$$\begin{aligned} &= X^{1/2-1/4}+aXY+(w/2)Y^2 \\ &+ r[(c_{40}/6)X(X^{1/2-1/4}) \\ &\quad + (c_{31}/6)(3X^{1/2-1/4})Y \\ &\quad + (c_{22}/2)XY^2 \\ &\quad + (c_{13}/6)Y^3], \end{aligned}$$

$F_Y(X,Y)$

$$\begin{aligned} &= sY+(a/2)(X^{1/2-1/4})+wXY+(z/2)Y^2 \\ &+ r[(c_{31}/6)X(X^{1/2-1/4}) \\ &\quad + (c_{22}/2)X^2Y \\ &\quad + (c_{13}/2)XY^2 \\ &\quad + (c_{04}/6)Y^3]. \end{aligned}$$

Let

$$\mathbf{S}=(1/2,0),$$

$$\mathbf{M}=(-1/2,0).$$

Define

$$F=|a|/8+r|c|^{31}/48,$$

$$\mathbf{F}_*=1+\mathbf{F},$$

$$\mu=KR^5,$$

$$\mathbf{W}=\mathbf{F}_*/\mu,$$

$$\sigma=4R\mathbf{W},$$

$$\rho=8R\mathbf{W},$$

$$\kappa=1/(16R),$$

$$C_{\text{cap}}(U)=2KR^5+2R.$$

Define the regular-jet condition

$$\text{REG}_r(U): rR^5 \leq \epsilon_0.$$

Define local saddle coordinates

$$\xi=1/2-X,$$

$$y=Y,$$

and the fixed cone

$$C_S=\{\xi \geq 0, \quad |y| \leq \kappa \xi\}.$$

Define the central strip and endpoint regions

$$S_{\text{strip}}=\{|X| \leq 1/2, \quad |Y| \leq W\},$$

$$S_{\text{center}}=\{|X| \leq 1/2-\sigma, \quad |Y| \leq W\},$$

$$B_S=B(S,\rho),$$

$$B_M=B(M,\rho).$$

THEOREM P02-LM-001.

Assume

$$\text{REG}_r(U)$$

and

$$s \leq -C_{\text{cap}}(U).$$

Then:

1. S and M are exact equilibria of $F_r^{(4)}$.

2. The Jacobian at S has one positive and one negative eigenvalue. The branch of the one-dimensional unstable manifold whose tangent points into increasing ξ lies strictly inside C_S .

3. $C_S \cap B_S$ is forward invariant until the branch reaches the section $\xi = \sigma$. The branch reaches that section before leaving B_S , and at the crossing

$$|Y| \leq \kappa \sigma = W/4 < W.$$

4. The strip S_{strip} is strictly forward invariant in its horizontal boundaries:

$$F_Y(X, W) \leq -1,$$

$$F_Y(X, -W) \geq 1$$

$$\text{for } |X| \leq 1/2.$$

5. In the central rectangle S_{center} ,

$$F_X(X, Y) \leq -3RW < 0.$$

Consequently every trajectory entering the right central section $X = 1/2 - \sigma$ with $|Y| < W$ crosses the central rectangle from right to left in finite forward time without leaving S_{strip} .

6. The matrix $-DF_r^{(4)}(M)$ is uniformly positive definite with

$$\lambda_{\min}(-DF_r^{(4)}(M)) \geq 63/128.$$

The ball B_M is forward invariant, contains no equilibrium other than M, and every trajectory entering B_M converges exponentially to M.

7. The left central section

$$X = -1/2 + \sigma,$$

$$|Y| \leq W$$

lies strictly inside B_M .

Therefore the selected increasing- ξ unstable branch of S enters S_{strip} , crosses the central region, enters B_M , and converges exponentially to M.

Equivalently, S and M are connected by the declared selected branch in the degree-four local flow.

3. HYPOTHESES AND PARAMETER SCOPE

The theorem is uniform over every U in R^8 and every $0 < r \leq 1$ satisfying $rR(U)^5 \leq \epsilon_0$ and $s \leq -C_{\text{cap}}(U)$.

The theorem permits equality in both hypothesis inequalities. Strict dynamical conclusions remain valid at the equality boundary because the derived cone, strip, transit, and sink margins remain strictly positive.

$R \geq 1$ automatically. The geometry is nondegenerate even when $U=0$ because $F_* = 1 + F \geq 1$ and hence $W > 0$.

No compact upper bound on R is assumed. Large R is compensated by the shrinking regularity domain $r \leq \epsilon_0/R^5$ and by the adaptive geometry.

4. EXPLICIT EXCLUSIONS

This closed lemma does not establish:

- any property of the full exact Gaussian field beyond its degree-four Taylor model;
- robustness under fifth-and-higher derivatives;
- probability under a Gaussian or Palm law;
- determinant-weighted mass of the hypothesis set;
- $1 - a_r = O(r^3)$;

- **P0.2**;

- **P0.1**;

- Theorem B;
- an accessible- r numerical guarantee outside REG_r ;
- optimality or necessity of C_{cap} ;
- any canonical theorem or machine-state promotion.

The theorem is sufficient, conditional, deterministic, local-model, and asymptotic-domain compatible.

5. DEPENDENCY MAP

D0 — Exact vector field and endpoint conventions.

Source: GP-DER-047 Section 2.

Status: CLOSED WITH THIS LEMMA.

D1 — Elementary adaptive-geometry inequalities.

Source: GP-DER-047 Lemma 3.1.

Status: CLOSED.

D2 — Strict transverse contraction and strip boundary flux.

Source: GP-DER-047 Lemma 4.1.

Status: CLOSED.

D3 — Central error and monotone transit.

Source: GP-DER-047 Lemmas 5.1–5.2.

Status: CLOSED.

D4 — Endpoint Jacobian variation.

Source: GP-DER-047 Lemma 6.1.

Status: CLOSED.

D5 — Fixed-coordinate saddle cone and selected unstable branch.

Source: GP-DER-047 Lemmas 7.1–7.3.

Status: CLOSED.

D6 — Sink contraction, invariant M ball, and central-to-sink geometry.

Source: GP-DER-047 Lemmas 8.1–8.3.

Status: CLOSED.

D7 — Composition into complete degree-four capture.

Source: GP-DER-047 Theorem 9.1.

Status: CLOSED.

The lemma has no probabilistic, exact-field, or external-theorem dependency.

6. PROOF ROADMAP

The proof uses one adaptive width W and two endpoint scales σ and ρ .

First, the threshold $s \leq -2\mu - 2R$ makes the Y -direction strongly contracting throughout the strip. Because $\mu W = F_*$, the horizontal strip boundary flux has a unit buffer rather than merely a nonpositive sign.

Second, the perturbation of $X^2 - 1/4$ is at most $0.501RW$. The central loss $\sigma(1 - \sigma)$ exceeds $3.99RW$, leaving $\dot{X} \leq -3RW$.

Third, the endpoint Jacobian changes by at most $3/64$ in each endpoint ball. At S , the large separation A - D generated by the threshold dominates the mixed term and nonlinear variation, yielding an invariant fixed cone with positive ξ growth. At M , the same threshold yields a strongly positive matrix $-J_M$, and the Jacobian variation is less than half its contraction margin.

Finally, the cone handoff identity $\kappa \sigma = W/4$ places the selected branch inside the strip, and the geometric identity $\sqrt{\sigma^2 + W^2} < \rho$ places the left central section inside the M ball. The components therefore glue without an uncovered transition region.

7. PROOF-OBLIGATION LEDGER

O1 — S and M are exact equilibria.

Disposition: PASS. Every pin-preserving polynomial factor vanishes at $X = \pm 1/2, Y = 0$.

O2 — Geometry is positive and lies in the intended chart.

Disposition: PASS. $W > 0, \sigma < 1/4, \rho < 1$.

O3 — Strict strip confinement.

Disposition: PASS. $\partial_Y F_Y \leq -\mu$ and boundary fluxes have at least unit inward margin.

O4 — Central monotone transit.

Disposition: PASS. $F_X \leq -3RW$.

O5 — Endpoint Jacobian variation.

Disposition: PASS. $\|D^2F\| \leq 12R$ and $12R\rho \leq 3/64$.

O6 — Saddle type and cone invariance.

Disposition: PASS. $\det DF(S) < 0$; cone constants exceed $7/8$ and $511R$ respectively.

O7 — Selected branch identity and handoff.

Disposition: PASS. The positive eigenvector slope is less than κ ; ξ grows; $\kappa\sigma = W/4$.

O8 — Sink type and contraction.

Disposition: PASS. $\lambda_{\min}(-DF(M)) \geq 63/128$ and nonlinear variation is below half the contraction margin.

O9 — Central section enters M ball.

Disposition: PASS. $\sqrt{\sigma^2 + W^2} < \rho$.

O10 — Global composition within the local polynomial flow.

Disposition: PASS. Cone $\rightarrow$ strip $\rightarrow$ transit $\rightarrow$ M-ball transitions cover the complete selected trajectory.

O11 — Equality and zero-jet edge cases.

Disposition: PASS. $U=0$, REG equality, and threshold equality retain positive strict dynamical margins.

O12 — Large-R behavior.

Disposition: PASS within scope. All inequalities remain valid for $R \geq 1$; the theorem domain shrinks as R^{-5} but is nonempty for every finite U .

8. INDEPENDENT REVIEW

Review artifact:

CL-AUD-049-v1.0.

Provider/model family: Anthropic/Claude.

Review type: ORGANIZATIONALLY DISTINCT / DERIVATION-BLIND ANALYTIC RECONSTRUCTION WITH SHARED-DEFINITION FIXTURE.

Original GP proof visible: NO.

Definitions supplied through a CL harness: YES.

Independent analytic route: YES.

Independent symbolic differentiation of endpoint Jacobians: YES.

Machine corroboration: 4,800 parameter points, zero violations.

Verdict: V1 DISCHARGED / all certificate inequalities independently derived on the declared domain.

Lineage limitation:

The reviewer received definitions through a same-family CL harness whose author had seen GP-DER-047. This could have propagated a definition transcription error. The current closure process separately compared the displayed field, endpoint Jacobians, geometry, and hypothesis definitions in GP-DER-047 against CL-AUD-049. No mismatch was found. CL-AUD-049 also differentiated the field independently, reducing the remaining shared-fixture risk to the explicit scalar geometry definitions, which were directly crosswalked.

The review qualifies under OP-PROT-011 as an organizationally distinct review of the exact conditional deterministic object. It counts once, not as multiple Claude reviews.

9. ADVERSARIAL TESTING

LCR-DATA-010 executed 200,000 adversarial parameter draws over the full REG-scaled hypothesis domain, including:

- - the exact $U=0$ edge;
- - exact REG-boundary equality;
- - threshold equality and near equality;
- - Gaussian coefficients;
- - one-hot coefficients spanning six to three positive logarithmic orders;
- - Cauchy heavy tails;
- - uniform wide coefficients;
- - extreme R above 10^5 .

No negative strict margin was found. The smallest sampled section-in-ball margin was positive but extremely small at very large R , as expected from the shrinking adaptive geometry. This does not threaten the exact algebraic inequality.

The source and receipt are marker-delimited and hash-reconstructable in Drive.

10. COUNTEREXAMPLE SEARCH AND FALSIFIERS

The lemma is false or must be reopened if any of the following is demonstrated under its exact hypotheses:

- - S or M is not an equilibrium;
- - the endpoint Jacobian formulas differ from the displayed derivatives;
- - a strip boundary has outward or zero flux;
- - F_X is nonnegative somewhere in S_{center} ;
- - the selected increasing- ξ unstable eigenvector lies outside C_S ;
- - the cone loses invariance before $\xi=\sigma$;
- - the branch leaves B_S before the handoff;
- - $-DF(M)$ fails the claimed eigenvalue lower bound;
- - B_M is not invariant or contains another equilibrium;
- - the left central section is not contained in B_M ;
- - a trajectory satisfying all hypotheses fails the stated branch composition;
- - CL-AUD-049's frozen definitions materially differ from GP-DER-047.

No such counterexample is currently present.

11. SOURCE AND VERSION HISTORY

Primary proof:

GP-DER-047-v1.0, created 2026-07-20/21.

Independent review:

CL-AUD-049-v1.0, created 2026-07-21.

Closure stress capsule:

LCR-DATA-010-v1.0, created 2026-07-24.

Terminal capsule:

LCR-CAP-009-v1.0, created 2026-07-24.

No mathematical statement was changed during closure. The closure extracts the deterministic Theorem 9.1 component from the broader unclosed P0.2 candidate chain and assigns it a unique lemma identity and exact scope.

12. CLOSURE DECISION

Closure ID: LCR-CLS-008-v1.0

Lemma ID: P02-LM-001

Date: 2026-07-24

Terminal label:

CLOSED — CONDITIONAL EXACT DEGREE-FOUR DETERMINISTIC CAPTURE MAJORANT.

Decisive evidence:

- - complete exact proof in GP-DER-047 Sections 2–9;
- - organizationally distinct derivation-blind analytic reconstruction in CL-AUD-049;
- - direct source-definition crosswalk;
- - reproducible hostile stress test with no counterexample;
- - explicit edge-case and scope analysis;
- - no unresolved critical, high, or medium mathematical objection within the frozen deterministic scope.

Closure authority:

OP-PROT-011 independent-eyes condition is satisfied by the Anthropic review. This terminal record closes the lemma as a reusable exact conditional deterministic result. It does not promote P0.2 or change the canonical mathematical program state.

13. DOWNSTREAM USAGE AUTHORIZED

P02-LM-001 may be used as the deterministic degree-four implication

$\text{REG}_r(U)$ and $q/r \leq -C_{\text{cap}}(U)$

$\Rightarrow$ selected degree-four S branch converges to M.

Any downstream use must preserve:

- - the exact field polynomial;
- - the exact U/R definitions;
- - $K=4096$ and $\epsilon_0=1/(1024K)$;
- - the REG condition;
- - the threshold $C_{\text{cap}}=2KR^5+2R$;
- - the branch orientation;
- - the conclusion's degree-four deterministic scope.

The lemma may serve as an input to a separately proved robust exact-field transfer lemma and a separately proved probability lemma.

It may not be cited alone as proof of exact-field adjacency or $O(r^3)$ probability.

14. REOPENING CONDITIONS

Reopen P02-LM-001 if:

- - an exact counterexample under the hypotheses appears;
- - a coefficient, sign, endpoint convention, or branch orientation is corrected;
- - the CL frozen-definition crosswalk is shown false;
- - a proof inequality is found to use a premise not included in the statement;
- - a dependency cycle or hidden exact-field assumption is discovered.

Do not reopen merely because:

- - better constants are found;
- - the theorem domain is impractically small at accessible r ;
- - a stronger exact-field theorem is desired;
- - another proof is developed;
- - P0.2 later fails at a probabilistic or transfer layer.

15. MANIFEST

P02-LM-001 primary proof

- - Artifact: GP-DER-047-v1.0
- - Drive ID: 1jm1WxNmDd_rBLZA9FQsc2Nlzb9_PPCuEBV0fX3mCL1c
- - Role: exact proof and definitions
- - Status: controlling primary proof for this lemma

P02-LM-001 independent review

- - Artifact: CL-AUD-049-v1.0
- - Drive ID: 1fzL4HgJDsawb6ew66XpOCfJGnnxw6TFTom7BM3H8OnI
- - Role: organizationally distinct derivation-blind reconstruction
- - Status: PASS with disclosed shared-definition fixture

P02-LM-001 hostile stress capsule

- - Artifact: LCR-DATA-010-v1.0
- - Drive ID: 1It6VYebSawBCA3S7Zs_g7ndB26-yr_NH6K_yNQs373Q
- - Role: source-identified numerical falsification aid
- - Status: PASS / zero independence credit

P02-LM-001 closure capsule

- - Artifact: LCR-CAP-009-v1.0
- - Drive ID: 1_ovqOV8q1YCC9J5Q72fYsfzGg3v_5Z71Uv4uwyZU3N4
- - Role: frozen statement, review package, closure record, and downstream fence
- - Status: terminal within exact conditional scope

16. REVIEW CHECKLIST RESULT

Statement exact: PASS.

Variables/domains explicit: PASS.

Quantifiers explicit: PASS.

Proof obligations discharged: PASS.

Cases and equality boundaries covered: PASS.

Dependencies noncircular: PASS.

Independent review version-specific: PASS.

Independence lineage disclosed: PASS.
Computational source identity: PASS.
Counterexample search meaningful: PASS.
Residual limitations explicit: PASS.
Downstream scope mapped: PASS.
P0.2 separation explicit: PASS.

FINAL VERDICT:

APPROVE AND CLOSE P02-LM-001 WITHIN THE EXACT CONDITIONAL DEGREE-FOUR DETERMINISTIC SCOPE.

END LCR-CAP-009-v1.0

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - CONDITIONAL EXACT DEGREE-FOUR DETERMINISTIC CAPTURE MAJORANT
Closure date	2026-07-24
Evidence basis	Complete exact proof, derivation-blind organizationally distinct review, definition crosswalk, and 200,000-sample hostile falsifier.
Independent review already recorded	Anthropic independently reconstructed the deterministic proof; shared-definition exposure was disclosed.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- No exact-field transfer, Gaussian/Palm mass, adjacency rate, P0.1/P0.2, or Theorem B.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____
Reproduction artifacts	Code/log/hash/location: _____

Signature and date _____

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: https://docs.google.com/document/d/1_ovqOV8q1YCC9J5Q72fYsfzGg3v_5Z71Uv4uwyZU3N4/edit

Closure / terminal record:

https://docs.google.com/document/d/1_ovqOV8q1YCC9J5Q72fYsfzGg3v_5Z71Uv4uwyZU3N4/edit

Local source snapshot used to construct this packet: LCR-CAP-009-v1.0 — P02-LM-001 Closed Conditional Deterministic Capture Lemma

Local source snapshot SHA-256 (DOCX export, transport checksum only):

4844d656fa98f5abaf36fe085f131295fc6f97c424d2de64fc8964a9451fe7b7

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.